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Cisco Cisco CleanAir

Report of Evidence of Use

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Patent Number: US7085289B2

1. A method for conserving power in a network device on a network, said method comprising: evaluating an address of an incoming message at the network device while the network device operates at a low bandwidth; determining whether said incoming message is addressed to the network device; and upon a determination that said incoming message is addressed to the network device, determining whether said incoming message is a ping packet or a data packet, wherein a bandwidth of the network device is increased to a higher bandwidth for receiving subsequent messages addressed to the network device if said incoming message is a data packet, and wherein the bandwidth of the network device is maintained at said low bandwidth if said incoming message is a ping packet.

Element 1:

1. A method for conserving power in a network device on a network, said method comprising:

Source: Cisco Wireless Controller Configuration Guide, Release 8.8

https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-8/config-guide/b_cg88/wireless_quality_of_service.html

Chapter: Wireless Quality of Service

CleanAir is supported on mesh AP backhaul at a 5-GHz radio of mesh. You can enable CleanAir on backhaul radios and can provide report interference details and air quality.

U-APSD

Unscheduled **automatic power save delivery (U-APSD)** is a QoS facility **defined in**

IEEE 802.11e that extends the battery life of mobile clients. In addition to extending battery life, this feature reduces the latency of traffic flow delivered over the wireless media. Because U-APSD does not require the client to poll each individual packet buffered at the access point, it allows delivery of multiple downlink packets by sending a single uplink trigger packet. U-APSD is enabled automatically when WMM is enabled.

Element 2:

evaluating an address of an incoming message at the network device

Source: Cisco Wireless Controller Configuration Guide, Release 8.8

https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-8/config-guide/b_cg88/wireless_quality_of_service.html

Chapter: Wireless Quality of Service

A Cisco CleanAir system consists of CleanAir-enabled access points, Cisco Wireless LAN Controllers, and Cisco Prime Infrastructure. These access **points collect information about all devices that operate in the industrial, scientific, and medical (ISM) bands,** identify and evaluate the information as a potential interference source, and forward it to the Cisco WLC. The Cisco WLC controls the access points, collects spectrum data, and forwards information to Cisco Prime Infrastructure or a Cisco mobility services engine (MSE) upon request.

Element 3:

while the network device operates at a low bandwidth;

Source: Cisco Wireless Controller Configuration Guide, Release 8.8

https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-8/config-guide/b_cg88/wireless_quality_of_service.html

Chapter: Wireless Quality of Service

Step 5

Ensure that any sources of interference that need to be detected and reported by the Cisco CleanAir system appear in the Interferences to Detect box and any that do not need to be detected appear in the Interferences to Ignore box. By default, all interference sources are detected, except BLE Beacon. The possible sources of interference that you can choose are as follows:

Spectrum 802.11 FH—An 802.11 frequency-hopping device (802.11b/g/n only)

Bluetooth Paging Inquiry—A Bluetooth discovery (802.11b/g/n only)

Bluetooth Sco Acl—A Bluetooth link (802.11b/g/n only)

Generic DECT—A digital enhanced cordless communication (DECT)-compatible phone

Generic TDD—A time division duplex (TDD) transmitter

Generic Waveform—A continuous transmitter

Jammer—A jamming device

Microwave—A microwave oven (802.11b/g/n only)

Canopy—A canopy bridge deviceSpectrum 802.11 inverted—A device using spectrally inverted Wi-Fi signals

Spectrum 802.11 non std channel—A device using nonstandard Wi-Fi channels

Spectrum 802.11 SuperG—An 802.11 SuperAG device

Spectrum 802.15.4—An 802.15.4 device (802.11b/g/n only)

Video Camera—An analog video camera

WiMAX Fixed—A WiMAX fixed device (802.11a/n/ac only)

WiMAX Mobile—A WiMAX mobile device (802.11a/n/ac only)

XBox—A Microsoft Xbox (802.11b/g/n only)

Note

APs that are associated to the Cisco WLC send interference reports only for the interferers that appear in the Interferences to Detect box. This functionality allows you to filter out interferers that you do not want as well as any that may be flooding the network and causing performance problems for the Cisco WLC or Prime Infrastructure. Filtering allows the system to resume normal performance levels.

Prior to Release 8.7, if CleanAir was enabled, by default, the BLE Beacon was included in the list of interferences to be detected. In 8.7 and later releases, the functionality is changed wherein the BLE Beacon is by default excluded from the list of interferences to be detected. This change in functionality is made to eliminate unwanted off-channel scans to improve performance. If you want the BLE Beacon to be detected, you will have to explicitly add it to the list of interferences to be detected.

Element 4:

determining whether said incoming message is addressed to the network device; and

Source: Cisco Wireless Controller Configuration Guide, Release 8.8

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Chapter: Wireless Quality of Service

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Spectrum 802.11 FH—An 802.11 frequency-hopping device (802.11b/g/n only)

Element 5:

upon a determination that said incoming message is addressed to the network device,

Step 1

Enable or disable VoIP snooping for a particular WLAN by entering this command:

config wlan call-snoop {enable | disable } wlan_id

Step 2

Save your changes by entering this command:

save config

Step 3

See the status of media session snooping on a particular WLAN by entering this command:

show wlan wlan_id

Information similar to the following appears:

WLAN Identifier..... 1

Profile Name..... wpa2-psk

Network Name (SSID)..... wpa2-psk

Status..... Enabled

...

FlexConnect Local Switching..... Disabled

FlexConnect Learn IP Address..... Enabled

Infrastructure MFP protection..... Enabled (Global Infrastructure MFP□Disabled)

Client MFP..... Optional

Tkip MIC Countermeasure Hold-down Timer..... 60

Call Snooping..... Enabled□

Element 6:

determining whether said incoming message is a ping packet or a data packet,

Source: Cisco Wireless Controller Configuration Guide, Release 8.8

https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-8/config-guide/b_cg88/wireless_quality_of_service.html

Chapter: Wireless Quality of Service

Step 8:

Choose 802.1p from the Protocol Type drop-down list and enter the maximum priority value in the 802.1p Tag text box to define the maximum value (0–7) for the priority tag associated with packets that fall within the profile.

The tagged packets include CAPWAP data packets (between access points and the controller) and packets sent toward the core network.

Note

If a QoS profile has 802.1p tagging configured and if this QoS profile is assigned to a WLAN that uses an untagged interface on the controller, the client traffic will be blocke...

Element 7:

wherein a bandwidth of the network device is increased to a higher bandwidth

Source: Cisco Wireless Controller Configuration Guide, Release 8.8

https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-8/config-guide/b_cg88/wireless_quality_of_service.html

Chapter: Wireless Quality of Service

Call Admission Control

Call admission control (CAC) enables an access point to maintain controlled quality of service (QoS) when the wireless LAN is experiencing congestion. The Wi-Fi Multimedia (WMM) protocol deployed in CCXv3 ensures sufficient QoS as long as the wireless LAN is not congested. However, in order to maintain QoS under differing network loads, CAC in CCXv4 is required. Two types of CAC are available: bandwidth-based CAC and load-based CAC.

Expedited Bandwidth Requests

The expedited bandwidth request feature enables CCXv5 clients to indicate the urgency of a WMM traffic specifications (TSPEC) request (for example, an e911 call) to the WLAN. When the controller receives this request, it attempts to facilitate the urgency of the call in any way possible without potentially altering the quality of other TSPEC calls that are in progress.

You can apply expedited bandwidth requests to both bandwidth-based and load-based CAC. Expedited bandwidth requests are disabled by default. When this feature is disabled, the controller ignores all expedited requests and processes TSPEC requests as normal TSPEC requests.

This table lists examples of TSPEC request handling for normal TSPEC requests and expedited bandwidth requests.

Element 8:

for receiving subsequent messages addressed to the network device

Element 9:

if said incoming message is a data packet, and

Source: Cisco Wireless Controller Configuration Guide, Release 8.8

https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-8/config-guide/b_cg88/wireless_quality_of_service.html

Chapter: Wireless Quality of Service

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The tagged packets include CAPWAP data packets (between access points and the controller) and packets sent toward the core network.

Note

If a QoS profile has 802.1p tagging configured and if this QoS profile is assigned to a WLAN that uses an untagged interface on the controller, the client traffic will be blocke...

Element 10:

wherein the bandwidth of the network device is maintained at said low bandwidth if said incoming message is a ping packet.

Source: Cisco Wireless Controller Configuration Guide, Release 8.8

https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-8/config-guide/b_cg88/wireless_quality_of_service.html

Chapter: Wireless Quality of Service

Spectrum 802.11 FH—An 802.11 frequency-hopping device (802.11b/g/n only)

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We understand that for the purpose of assisting you in collecting data, you may disclose to us certain confidential information that you want to keep confidential; therefore, for good and valuable consideration, the receipt and sufficiency of which is hereby acknowledged, we agree as follows:

1. For a period of twelve (36) months from the date we receive your confidential information ("Effective Date"), we will not use or disclose the information, except: (a) in the event that such information (i) comes into the public domain not through us; (ii) is already known to us at the time of disclosure; (iii) is lawfully otherwise learnt or independently developed by us; or (iv) is required to be disclosed by law.

2. Since a breach of our obligations hereunder may result in irreparable harm to you, we agree that upon such breach or threat thereof, you shall be entitled to specific performance and other injunctive relief without posting a bond, and in the event of disclosures required by court order, we shall promptly notify you of the proceeding.

3. This agreement supersedes all prior discussions and writings and constitutes the entire agreement between you and us with respect to the subject matter hereof. This Agreement shall be binding upon the successors and assigns of the respective parties, and may not be modified except by an instrument executed by both parties.

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